The two hyperbolic bounds for the Kilbas–Saigo function

MathDB 348998: proof throughout the conjectured parameter range

Principia Math In Progress

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Task 1: the full conjecture

For $\alpha, m > 0$ and $l > -1/\alpha$, write

$$E_{\alpha,m,l}(z) = 1 + \sum_{n=1}^{\infty} \left(\prod_{k=1}^{n} \frac{\Gamma(1+\alpha((k-1)m+l))}{\Gamma(1+\alpha((k-1)m+l+1))} \right) z^n. \tag{1}$$

Theorem 1. *Let $0 < \alpha \le 1$, $m > 0$, and $l > m - 1/\alpha$. Define*

$$c_- = \frac{\Gamma(1+\alpha(l-m))}{\Gamma(1+\alpha(l-m+1))}, \qquad c_+ = \frac{\Gamma(1+\alpha l)}{\Gamma(1+\alpha(1+l))}.$$

Then, for every $x \ge 0$,

$$\boxed{\frac{1}{1+c_- x} \;\le\; E_{\alpha,m,l}(-x) \;\le\; \frac{1}{1+c_+ x}.} \tag{2}$$

Both inequalities are strict when $x > 0$, and both are equalities at $x = 0$. The two constants are optimal within bounds of these forms.

This is Conjecture 4 in the published paper [1], numbered Conjecture 4.5 in its arXiv version. We prove both bounds with exactly the printed hypotheses. The proof uses classical facts about subordinators and their exponential functionals, credited to [2]; the needed moment and comparison arguments are given below.

A positive gamma-ratio exponent

Set

$$b = 1 + \alpha(l-m) > 0, \qquad d = \alpha m > 0, \qquad \phi(s) = \frac{\Gamma(b+ds+\alpha)}{\Gamma(b+ds)}, \quad s \ge 0. \tag{3}$$

Thus $c_- = 1/\phi(0)$ and $c_+ = 1/\phi(1)$. For $0 < \alpha < 1$ the beta integral and integration by parts give

$$\phi(s) = q + \frac{\alpha}{\Gamma(1-\alpha)} \int_0^\infty (1 - e^{-dst}) \frac{e^{-(b+\alpha)t}}{(1-e^{-t})^{1+\alpha}}\, dt, \qquad q = \phi(0) > 0. \tag{4}$$

Here is a direct derivation. For $v > 0$,

$$\frac{\Gamma(v+\alpha)}{\Gamma(v)} = \frac{v}{\Gamma(1-\alpha)} \int_0^\infty e^{-vt}(e^t-1)^{-\alpha}\, dt = \frac{\alpha}{\Gamma(1-\alpha)} \int_0^\infty (1-e^{-vt}) \frac{e^{-\alpha t}}{(1-e^{-t})^{1+\alpha}}\, dt.$$

For the integration by parts, the boundary product $(1-e^{-vt})(e^t-1)^{-\alpha}$ tends to zero at both ends: it is $O(t^{1-\alpha})$ at zero and decays at infinity. Subtract the last identity at $v = b$ from that at $v = b + ds$ to obtain (4). All integrals so subtracted are finite.

The measure in (4) has density of order $t^{-1-\alpha}$ at zero and exponential decay at infinity. After replacing the jump size t by dt, it satisfies the subordinator integrability condition $\int (1 \wedge y)\, \nu(dy) < \infty$. Consequently there exists a nondecreasing Lévy process S, starting at zero, such that

$$\mathbb{E}e^{-sS_t} = e^{-t(\phi(s)-q)}, \qquad s, t \geq 0. \tag{5}$$

This is the standard subordinator construction from its positive jump measure; it also follows by truncating the small jumps of a Poisson random measure and taking the increasing limit. When $\alpha = 1$, the same conclusion holds with $\phi(s) = b + ds$, $q = b$, and the deterministic process $S_t = dt$. Thus the endpoint $\alpha = 1$ is covered separately and exactly.

Identifying the exponential functional

Let ζ be independent of S, with exponential distribution of rate q, and define

$$I = \int_0^\zeta e^{-S_t}\, dt.$$

In particular $0 < I \leq \zeta$ almost surely. For each integer $n \geq 1$, Tonelli's theorem and independent increments yield the classical moment formula

$$\mathbb{E}I^n = \frac{n!}{\prod_{k=1}^n \phi(k)}. \tag{6}$$

For completeness, expand the nth power as $n!$ times the integral over $0 < t_1 < \cdots < t_n < \zeta$. Put $t_0 = 0$ and $u_j = t_j - t_{j-1}$. The expectation of the integrand, including the survival probability e^{-qt_n}, is

$$\exp\left(-\sum_{j=1}^n u_j \phi(n-j+1)\right).$$

Integration over all positive u_j proves (6). This is also [2, Theorem 2(i)].

The function $\phi(s)$ increases to infinity. For $\alpha < 1$ this follows by monotone convergence in (4), since the jump measure has infinite total mass; for $\alpha = 1$ it is immediate. Hence $\sum_{n\geq 0} r^n / \prod_{k=1}^n \phi(k)$ converges for every $r \geq 0$ by the ratio test. Applying Tonelli to the exponential series and then dominated convergence for complex arguments proves

$$\mathbb{E}e^{zI} = \sum_{n=0}^\infty \frac{z^n}{\prod_{k=1}^n \phi(k)} = E_{\alpha,m,l}(z), \qquad z \in \mathbb{C}. \tag{7}$$

The last equality uses $b + dk = 1 + \alpha((k-1)m + l)$, so it identifies every coefficient in (1), not merely an asymptotic expansion. In particular

$$L(x) := E_{\alpha,m,l}(-x) = \mathbb{E}e^{-xI}, \qquad \mu := \mathbb{E}I = \frac{1}{\phi(1)}.$$

The lower bound

The pathwise inequality $I \leq \zeta$ gives, for $x \geq 0$,

$$L(x) \geq \mathbb{E}e^{-x\zeta} = \frac{q}{q+x} = \frac{1}{1+c_- x}. \tag{8}$$

The upper bound from the remaining lifetime

For $y > 0$ let

$$A_t = \int_0^t e^{-S_s}\, ds, \qquad T_y = \inf\{t \geq 0 : A_t > y\}.$$

The process A is continuous and strictly increasing at finite times. On $\{T_y < \infty\}$ one has $A_{T_y} = y$ and $T_y \geq y$. Also $\{I > y\} = \{T_y < \zeta\}$ up to null events. By the strong Markov property of S and the memorylessness of the independent killing time, the integral remaining after T_y, on this event, has the form

$$I - y = e^{-S_{T_y}} I',$$

where the future functional I' has the same law as I and is independent of the past under the surviving conditional law. Equivalently, without conditional notation,

$$\begin{aligned} \mathbb{E}(I-y)_+ &= \mu\, \mathbb{E}\Big[e^{-S_{T_y}-qT_y};\ T_y < \infty\Big] \\ &\leq \mu\, \mathbb{E}\big[e^{-qT_y};\ T_y < \infty\big] = \mu\, \mathbb{P}(I > y). \end{aligned} \tag{9}$$

The factor e^{-qT_y} explicitly accounts for survival up to the stopping time; it must not be omitted.

For $x > 0$, integrate (9) against $e^{-xy}\, dy$. Tonelli applies to the nonnegative integrands. Direct integration gives

$$\int_0^\infty e^{-xy}\mathbb{E}(I-y)_+\, dy = \frac{\mu}{x} - \frac{1-L(x)}{x^2}, \qquad \int_0^\infty e^{-xy}\mathbb{P}(I > y)\, dy = \frac{1-L(x)}{x}.$$

It follows that

$$\mu x - (1 - L(x)) \leq \mu x(1 - L(x)), \qquad \text{and thus} \qquad L(x) \leq \frac{1}{1+\mu x} = \frac{1}{1+c_+ x}.$$

Together with (8) and $L(0) = 1$, this proves both conjectured inequalities throughout the stated range.

Strictness, sharp constants, and an independent check

For every fixed $t > 0$, $S_t > 0$ almost surely. For $\alpha = 1$ this is deterministic. For $\alpha < 1$ it follows from

$$\mathbb{P}(S_t = 0) = \lim_{s\to\infty} \mathbb{E}e^{-sS_t} = \lim_{s\to\infty} e^{-t(\phi(s)-q)} = 0.$$

Monotonicity of the paths and a countable set of positive rational times give $S_t > 0$ simultaneously for all $t > 0$, almost surely. Therefore $I < \zeta$ almost surely and (8) is strict for $x > 0$. Since $I > 0$ almost surely, there is an $\varepsilon > 0$ such that $\mathbb{P}(I > \varepsilon) > 0$. For every $0 < y < \varepsilon$, $T_y \geq y$ makes the inequality in (9) strict. Its integral is then strict, proving the asserted upper strictness.

For the constants, (1) gives $L(x) = 1 - c_+ x + O(x^2)$ as $x \downarrow 0$. A uniform upper bound $L(x) \leq (1+cx)^{-1}$ therefore requires $c \leq c_+$. The established asymptotic in [1, Proposition 6] is $L(x) \sim q/x$ as $x \to \infty$. A uniform lower bound $(1+cx)^{-1} \leq L(x)$ consequently requires $c \geq 1/q = c_-$. This credits the source's already proved asymptotic; it is not a new asymptotic claim here.

There is also a short independent check of the upper bound. The classical factorization [2, Theorem 2(ii)] supplies a positive R, independent of I, with IR standard exponential and $\mathbb{E}R = \phi(1)$. Conditional Jensen gives

$$L(x) \leq \mathbb{E}\exp\left(-\frac{xIR}{\phi(1)}\right) = \frac{1}{1+x/\phi(1)}.$$

The preceding remaining-lifetime proof establishes the bound without relying on that factorization theorem.

Verification and source scope

The verifier passes thirteen exact symbolic identities, including parameter reindexing, the kernel integration by parts, the residual Laplace calculation, and six beta-law moments at $\alpha = 1$. Ninety-six finite series diagnostics cover 24 parameter triples at 160 and 220 digits. Twenty-four independent integrals check the explicit $\alpha = 1$ representation $I = (1 - e^{-d\zeta})/d$, and four quadratures check (4) at 100 digits. All checks pass. Numerical tests are supplementary; the proof above applies to every allowed parameter and every $x \geq 0$.

The complete published source and its arXiv version retain the conjecture. Dated bounded claim searches and a literature audit accompany this note. Classical probability inputs are explicitly credited. The authoring agent performed the audit; no independent peer review is claimed.

Tasks 2–5: proposed follow-up work

These are four proposed tasks, not additional solved problems.

2. **Task 2.** Prove the conjectured sharp lower bound with power $1 + 1/m$ at the boundary $l = m - 1/\alpha$, stated as Conjecture 3 in [1], for every $m > 0$.

3. **Task 3.** Determine which changes of α, m, l order the normalized variables $I/\mathbb{E}I$ in convex order, and derive the corresponding parameter monotonicity of $E_{\alpha,m,l}(-x/\mathbb{E}I)$.

4. **Task 4.** Construct sharp rational refinements of (2) that match the coefficient of x^2 at zero and the leading q/x term at infinity throughout the same range.

5. **Task 5.** Characterize the parameters for which the survival function of I is log-concave, strengthening the remaining-mean inequality (9) to an increasing hazard rate.

References

[1] L. Boudabsa and T. Simon, *Some Properties of the Kilbas–Saigo Function*, Mathematics **9**(3), 217 (2021), 24 pp., doi:10.3390/math9030217. Conjecture 4 and Proposition 6, p. 16. Preprint: arXiv:2012.05666, Conjecture 4.5 and Proposition 4.3, p. 17.

[2] J. Bertoin and M. Yor, *Exponential functionals of Lévy processes*, Probability Surveys **2** (2005), 191–212, doi:10.1214/154957805100000122. See §3.1 and Theorem 2, pp. 193–196. Complete journal reprint: arXiv:math/0511265.